W04 Foundation Programs Design

Enzo Torricella

This document defines the classes, private member variables, public behaviors, relationships, and execution flow for the YouTube Videos and Online Ordering programs.

Program 1 YouTube Videos

The program stores several videos and their comments, then displays each video's details and the number and text of its comments.

Video
- _title : string - _author : string - _length : int - _comments : List<Comment>
+ Video(title : string, author : string, length : int) + AddComment(comment : Comment) : void + GetNumberOfComments() : int + GetDisplayText() : string

Video 1 contains 0..* Comment

Comment
- _commenterName : string - _text : string
+ Comment(commenterName : string, text : string) + GetDisplayText() : string

Program flow

Program creates 3 Video objects -> adds 3 or 4 Comment objects to each video -> stores the videos in a list -> loops through the list -> calls GetDisplayText -> GetDisplayText calls GetNumberOfComments and formats every comment.

Program 2 Online Ordering

The program creates domestic and international orders, produces packing and shipping labels, and calculates the product total plus the correct one-time shipping charge.

Order
- _products : List<Product> - _customer : Customer
+ Order(products : List<Product>, customer : Customer) + CalculateTotalCost() : decimal + GetPackingLabel() : string + GetShippingLabel() : string

Order 1 contains 2..* Product **Order 1 belongs to 1 Customer**

Product
- _name : string - _productId : string - _price : decimal - _quantity : int
+ Product(name : string, productId : string, price : decimal, quantity : int) + CalculateTotalCost() : decimal + GetPackingLabelText() : string

Customer
- _name : string - _address : Address
+ Customer(name : string, address : Address) + IsInUSA() : bool + GetShippingLabelText() : string

Customer 1 has 1 Address

Address
- _street : string - _city : string - _stateOrProvince : string - _country : string
+ Address(street : string, city : string, stateOrProvince : string, country : string) + IsInUSA() : bool + GetFullAddress() : string

Program flow

Program creates Address, Customer, Product, and Order objects -> prints the packing label -> prints the customer shipping label -> sums Product.CalculateTotalCost results -> calls Customer.IsInUSA -> Customer calls Address.IsInUSA -> adds $5 domestic shipping or $35 international shipping -> prints the final order total.